

AI Engineer

Objective

Design a basic AI Copilot prototype that answers user queries from structured data stored in a NoSQL database (MongoDB). The data spans different CRM modules like Call Logs, Appointments, Email Conversations, and WhatsApp Business Conversations.

Problem Statement

You're building a CRM AI Copilot that can help users extract useful business insights from large volumes of interaction data. Your task is to write a simple backend or CLI interface that fetches and filters relevant information based on structured data from MongoDB collections.

Each module has its own dataset (500+ records). You'll work on writing aggregation logic and query patterns that can answer real-world business questions.

[Use this sample data for loading into the mongodb](#)

Module 1: Call Logs

Sample Record:

```
{
  "call_id": "CL1234",
  "agent": "Priya Sharma",
  "customer": "Rahul Jain",
  "call_time": "2024-03-15T10:00:00Z",
  "duration_minutes": 5,
  "call_status": "completed"
}
```

Tasks:

- Load into MongoDB

- Expose CLI/API to answer:
 - How many calls did Priya Sharma make this week?
 - List all failed calls in the last 7 days.
 - What is the average call duration for completed calls?
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Module 2: Appointments

Sample Record:

```
{  
  "appointment_id": "APPT1001",  
  "agent": "Neha Kapoor",  
  "customer": "Sneha Rao",  
  "appointment_time": "2024-03-20T14:00:00Z",  
  "status": "confirmed",  
  "notes": "Follow-up on demo"  
}
```

Tasks:

- Load into MongoDB
 - Expose CLI/API to answer:
 - List all confirmed appointments for today.
 - How many appointments has Neha Kapoor had this week?
 - Find missed appointments in the last 7 days.
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Module 3: Email Conversations

Sample Record:

```
{
  "email_id": "EML7890",
  "sender": "support@crm.io",
  "receiver": "customer@example.com",
  "subject": "Onboarding Help",
  "timestamp": "2024-03-22T09:30:00Z",
  "status": "sent"
}
```

Tasks:

- Load into MongoDB
- Expose CLI/API to answer:
 - List all emails sent by support@crm.io this week.
 - How many emails had 'onboarding' in the subject last month?
 - Show emails that were not delivered successfully.

Module 4: WhatsApp Business Conversations

Sample Record:

```
{
  "message_id": "WA5678",
  "agent": "Amit Verma",
  "customer": "Vikram Das",
  "timestamp": "2024-03-24T18:45:00Z",
  "message": "Please share your requirements.",
  "status": "delivered"
}
```

Tasks:

- Load into MongoDB
 - Expose CLI/API to answer:
 - List all WhatsApp messages sent by Amit Verma today.
 - How many messages were delivered vs failed in the last 3 days?
 - Find the latest message sent to Vikram Das.
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Deliverables

- Develop a basic AI Copilot prototype using the provided CRM datasets.
 - Load/use the sample data for:
 - Call Logs
 - Appointments
 - Email Conversations
 - WhatsApp Business Conversations
 - Expose a simple CLI/API to answer all the questions mentioned under each module.
 - Ensure responses are accurate, concise, and based on the underlying data.
 - Add a README with setup instructions, supported queries, approach, and sample inputs/outputs.
 - Share the code in a private repository and add psvamsi as collaborator.
 - A basic low-level flow diagram is preferred.
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Evaluation Criteria

- AI Output Quality - Accuracy, relevance, clarity, and usefulness of responses.
- Query Understanding - Correctly identify intent, entities, filters, and date ranges from user queries.
- Data Grounding - Responses should be derived from the available data without hallucination.
- Use Cases Covered - All specified queries across the four modules should be supported.
- Natural Language Handling - Ability to handle variations of the given queries.
- Code Quality - Clean, readable, modular implementation with appropriate error handling.
- Approach & Documentation - Clear explanation of the AI/query approach and key design decisions.